

PENRHYN, Cook Islands

WMO: 918020

Lat: 9.017S

Lon: 158.031W

Elev: 1

StdP: 101.31

Time Zone: -10.00 (W10)

Period: 86-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.5	25.0	20.2	14.9	27.6	20.9	15.6	27.8	13.0	26.4	11.3	26.9	3.8	80	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	3.3	32.8	26.9	32.3	26.7	31.9	26.6	27.8	31.2	27.6	30.9	27.3	30.5	2.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.2	26.6	22.2	29.9	26.3	21.7	29.6	89.0	31.5	87.6	31.0	86.4	30.8	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	10.1	9.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	28.4	28.3	28.6	28.8	28.8	28.6	28.4	28.1	28.2	28.4	28.4	
	DBStd	0.98	0.83	1.06	1.13	0.92	0.92	0.83	0.90	0.83	0.91	0.92	0.98	1.21
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6728	571	513	577	563	582	557	570	561	547	563	552	572
	CDD18.3	3686	313	280	318	313	323	307	312	303	297	305	302	313
	CDH23.3	43403	3703	3281	3819	3817	4004	3636	3579	3395	3404	3534	3570	3662
CDH26.7	14882	1271	1106	1411	1448	1546	1278	1153	988	1068	1124	1216	1274	
Wind	WSAvg	5.0	5.4	5.3	4.4	4.6	5.6	5.5	4.7	5.1	5.6	5.4	4.2	4.1
Precipitation	PrecAvg	1430	199	184	180	104	90	84	93	82	63	98	97	156
	PrecMax	2513	494	690	391	367	235	250	266	267	131	194	288	451
	PrecMin	514	44	27	39	33	14	20	16	14	10	12	21	12
	PrecStd	471	128	158	102	71	57	51	70	65	34	46	61	125
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	32.7	32.7	32.7	33.2	32.3	32.5	32.2	32.6	33.0	32.9	33.2
		MCWB	26.6	27.3	27.6	27.4	26.9	26.7	26.4	26.2	26.2	27.2	26.4	26.8
	2%	DB	31.8	31.8	32.0	32.0	32.1	31.6	31.7	31.5	31.6	31.9	32.0	32.3
		MCWB	26.2	27.0	27.3	27.1	26.7	26.4	26.1	25.9	25.9	26.2	26.3	26.7
	5%	DB	31.0	31.0	31.2	31.2	31.2	30.9	30.9	30.7	30.9	31.0	31.2	31.7
		MCWB	26.1	26.6	27.0	26.9	26.5	26.3	25.8	25.5	25.6	25.9	26.0	26.6
	10%	DB	30.2	30.2	30.5	30.5	30.4	30.2	30.1	29.9	30.1	30.1	30.5	30.9
		MCWB	26.0	26.3	26.6	26.7	26.3	26.0	25.6	25.3	25.5	25.7	25.9	26.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	28.0	28.2	28.2	27.8	27.6	27.3	27.1	27.2	27.6	27.7	27.8
		MCDB	31.0	31.8	31.7	31.2	30.9	30.8	31.1	30.1	30.3	31.7	30.9	31.7
	2%	WB	27.1	27.4	27.7	27.7	27.4	27.1	26.7	26.7	26.7	27.0	27.2	27.2
		MCDB	30.2	30.9	31.0	30.7	30.4	30.2	30.3	29.6	29.9	30.6	30.4	31.0
	5%	WB	26.7	27.0	27.3	27.4	27.1	26.8	26.4	26.3	26.4	26.6	26.7	26.9
		MCDB	29.7	30.3	30.5	30.4	30.1	29.8	29.7	29.3	29.4	29.9	29.9	30.6
	10%	WB	26.4	26.6	27.0	27.1	26.8	26.5	26.1	26.1	26.1	26.2	26.4	26.6
		MCDB	29.4	29.6	30.0	30.0	29.8	29.4	29.2	28.9	29.1	29.2	29.5	30.0
Mean Daily Temperature Range	5% DB	MDBR	3.4	3.5	3.6	3.4	3.3	3.5	3.5	3.5	3.7	3.4	3.6	4.0
		MCDBR	5.2	4.8	4.8	4.6	4.6	4.5	5.0	4.9	5.2	5.2	5.1	5.4
		MCWBR	2.4	2.2	2.4	2.2	2.1	2.1	2.3	2.3	2.2	2.2	2.2	2.3
	5% WB	MCDBR	3.7	4.1	4.1	3.7	3.5	3.7	3.9	3.7	3.8	4.0	3.8	4.6
		MCWBR	2.1	2.2	2.3	2.2	2.1	2.1	2.0	2.2	2.2	2.2	2.1	2.4
Clear-Sky Solar Irradiance	taub		0.383	0.386	0.381	0.384	0.384	0.386	0.384	0.381	0.390	0.379	0.378	0.382
	taud		2.625	2.615	2.633	2.635	2.634	2.604	2.597	2.596	2.568	2.606	2.621	2.615
	Ebn at Noon		957	950	939	908	880	861	871	898	916	945	955	956
	Edh at Noon		101	102	99	94	90	90	92	96	104	102	101	102
All-Sky Solar Radiation	RadAvg		6.04	6.10	5.97	5.83	5.31	5.06	5.29	5.92	6.48	6.59	6.59	6.33
	RadStd		0.60	0.78	0.52	0.41	0.30	0.30	0.38	0.29	0.30	0.38	0.40	0.67

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page